

RECAP: Fix a prob. space $(\Omega, \mathcal{F}, \mathbb{P})$

Consider a Time Homogeneous DTMC $\{X_n : n \geq 0\}$ on the state space S with transition probability matrix P . $P_{ij} = \mathbb{P}\{X_{n+1} = j | X_n = i\}$.

1) Stopping time - definition

2) Strong Markov property of a TH DTMC.

3) Hitting Times and Recurrence

- First passage time distribution: $f_{ij}^{(n)}$
 - Mean recurrence time: $v_{jj} = \sum_{n=1}^{\infty} n \cdot f_{jj}^{(n)}$

Consider a TH DTMC $\{X_n, n \geq 0\}$ on a state space S , defined w.r.t $(\Omega, \mathcal{F}, \mathbb{P})$.

For $i, j \in S$, $f_{ij}^{(n)} = \mathbb{P}\{X_1 \neq j, X_2 \neq j, \dots, X_{n-1} \neq j, X_n = j | X_0 = i\}$ prob. that first visit to state j is at time n when process starts in i

$f_{ij} = \sum_{n=1}^{\infty} f_{ij}^{(n)} \rightarrow$ prob. of ever hitting j when process starts off in i .

$f_{ij}^{(n)}$ is a pmf on $\mathbb{N}$, can be defective, in general.

First passage time distribution for hitting j starting from i .

$f_{jj}^{(n)}$ - prob. of first return to state j at time n

if $f_{jj} = 1$, Mean recurrence time $v_{jj} = \sum_{n=1}^{\infty} n f_{jj}^{(n)}$

- PLAN:
- 1) Convergence of Expectation & Monotone Convergence Theorem
 - 2) Transient and Recurrent States
 - 3) Communicating Classes and Class Properties
 - 4) Positive Recurrence & Stationary Distribution

REFERENCE: 1) A. Kumar, "Discrete Event Stochastic Processes"
 2) Lecture Notes of the course "Stochastic Processes", by Dr. P.N. Karthik, IITH.

Convergence of Expectation:

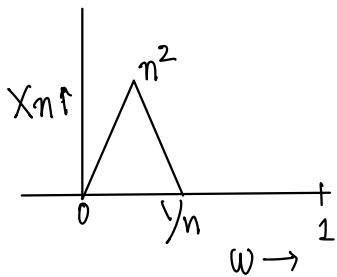
Ref: A. Kumar, "Discrete Event Stochastic Processes"

If $X_1, X_2, \dots$ is a sequence of random variables converging almost surely to the r.v. X , is it true that $\lim_{n \rightarrow \infty} \mathbb{E}[X_n] = \mathbb{E}\left[\lim_{n \rightarrow \infty} X_n\right]$? *Validity of the interchange of limit & expectation operators*

i.e., if $X_n \xrightarrow{a.s.} X$, is $\lim_{n \rightarrow \infty} \mathbb{E}[X_n] = \mathbb{E}[X]$?

In general, such an exchange is not valid.

Example:



$\Omega = [0, 1]$, $\mathcal{F} = \mathcal{B}([0, 1])$, $P = \text{unif prob. measure}$.

For each $0 < \omega \leq 1$, $\lim_{n \rightarrow \infty} X_n(\omega) = 0$. Define $X = 0 \ \forall \ \omega \in [0, 1]$.
 $X_n \xrightarrow{a.s.} X$.

$$\mathbb{E}[X_n] = \frac{1}{2} \times \frac{1}{n} \times n^2 = \frac{n}{2} \quad \lim_{n \rightarrow \infty} \mathbb{E}[X_n] = \infty \neq \underbrace{\mathbb{E}[X]}_0.$$

The following two results provide conditions that ensure the validity of the exchange of expectation and almost-sure limit of random variables.

Monotone Convergence Theorem: If $X_n, n \geq 1$ is a sequence of random variables that converges almost surely to the random variable X , i.e., $X_n \xrightarrow{a.s.} X$, and the following two conditions hold:

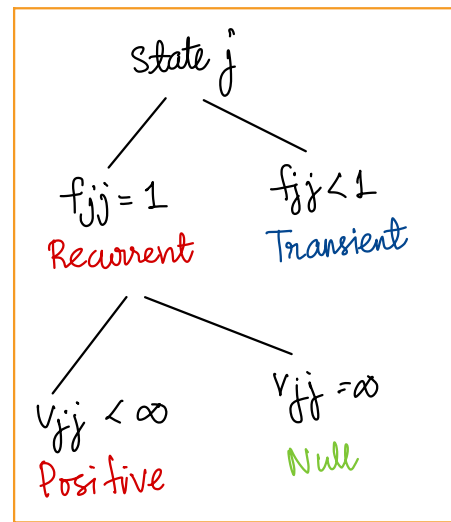
- (i) X_n is non-negative for every $n \geq 1$, and
- (ii) $X_1 \leq X_2 \leq \dots \leq X_n \leq \dots$ (sequence of RVs is monotonically non-decreasing)

$$(\Rightarrow \forall \omega \in \Omega, X_1(\omega) \leq X_2(\omega) \leq \dots \leq X_n(\omega) \leq \dots)$$

Then, $\lim_{n \rightarrow \infty} \mathbb{E}(X_n) = \mathbb{E}(X)$ { $\mathbb{E}(X) = \infty$ is a possibility here }.

Transient & Recurrent States

For a TH DTMC $\{X_n, n \geq 0\}$ on a state space S ,
a state $j \in S$ is called **transient**, if $f_{jj} < 1$,
recurrent, if $f_{jj} = 1$, **positive recurrent** if $f_{jj} = 1$ and
the mean recurrence time v_{jj} is finite, and **null-recurrent**
if $f_{jj} = 1$ and the mean recurrence time v_{jj} is infinite.



Number of Returns to a State

Let $M_j = \sum_{n=1}^{\infty} I_{\{X_n=j\}}$ - Number of visits to state j
for $n \geq 1$.

$$\begin{aligned} \mathbb{E}[M_j | X_0 = j] &= \mathbb{E}\left[\sum_{n=1}^{\infty} I_{\{X_n=j\}} \mid X_0 = j\right] \\ &= \sum_{n=1}^{\infty} p_{jj}^{(n)} \quad \left(\text{Using Monotone Convergence Theorem.}\right) \end{aligned}$$

How? (Homework)

Theorem (2.3 in Ref).

For a time homogeneous DTMC $\{X_n, n \geq 0\}$ on the state space S , if the state $j \in S$ is **transient**
($f_{jj} < 1$) then, for $m \in \{0, 1, 2, \dots\}$

$$\mathbb{P}(M_j = m \mid X_0 = j) = f_{jj}^m (1 - f_{jj})$$

(Proof: Homework!)

and if j is **recurrent** ($f_{jj} = 1$) then, for $m \in \{0, 1, 2, \dots\}$

$$\mathbb{P}(M_j = m \mid X_0 = j) = 0$$

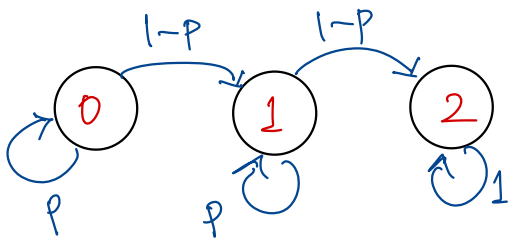
Using this result, if j is transient $\mathbb{E}[M_j \mid X_0 = j] = \sum_{m=1}^{\infty} m f_{jj}^m (1 - f_{jj})$
 $= \frac{f_{jj}}{1 - f_{jj}} < \infty$

And if j is recurrent $\mathbb{E}[M_j \mid X_0 = j] = \infty$

Theorem (2.4 in Ref).

- i) A state j is transient iff the expected number of returns is finite.
Thus, for a recurrent state j , the expected number of returns to j is infinite
i.e., $\sum_{n=1}^{\infty} p_{jj}^{(n)} = \infty$
- ii) For a transient state j , $\lim_{n \rightarrow \infty} p_{jj}^{(n)} = 0$.

Example.



Which states are transient?

Which are positive & null recurrent?

$$f_{00} = p, f_{11} = p, f_{22} = 1$$

$$v_{22} = \sum_{n=1}^{\infty} n f_{22}^{(n)} = 1 \cdot f_{22}^{(1)} = 1 < \infty$$

Communicating Classes and Irreducibility of a DTMC

Defn: Given a TH DTMC $\{X_n, n \geq 0\}$ on the state space S , for $i, j \in S$, we say that

- i) j is **reachable** from i if $\exists n \geq 0$ s.t. $p_{ij}^{(n)} > 0$ (Notation: $i \rightarrow j$)
($\exists n \geq 0$ s.t. j can be reached from i in n steps with positive probability)
- ii) i and j **communicate** if $i \rightarrow j$ and $j \rightarrow i$ (Notation: $i \leftrightarrow j$)

Question: Prove that the relation $\leftrightarrow$ on S is an equivalence.

- i) Reflexive:
- ii) Symmetric:
- iii) Transitive:

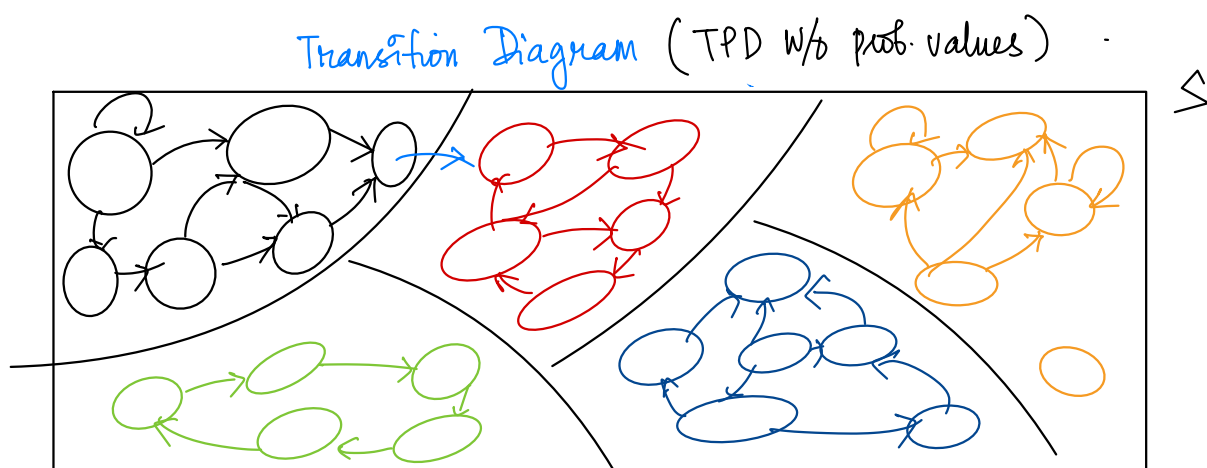
$\therefore \leftrightarrow$ partitions S into equivalence classes.

Defn: Given a TH DTMC $\{X_n, n \geq 0\}$ on the state space S with TPM P .

- i) The classes into which $\longleftrightarrow$ partitions S are called **communicating classes**.
- ii) A communicating class C is said to be **closed**, if $\nexists i \in C$ and $j \notin C, p_{ij} > 0$ } otherwise called OPEN
- iii) The Markov chain is said to be **irreducible** if the entire S is one communicating class.

Remark: i) An irreducible DTMC is closed.

ii) The communicating classes depend only on the position of the $\neq$ elements of the TPM, not on their actual values.



If $\nexists$ an edge that leaves a class C : OPEN
 $\nexists$ an edge leaving the class : CLOSED.

Irreducibility is also referred to as a **unichain** property.

Class Property Theorem: The states in a communicating class are either

- i) all transient,
- ii) all null-recurrent, or
- iii) all positive-recurrent.

(Without Proof).

Invariant (Stationary) Distribution (Also called steady state or equilibrium dist)

Consider a TH DTMC with a state space S and TPM P .

A PMF π on S is called an **invariant distribution** for P if $\pi = \pi P$

i.e. for all $j \in S$, $\pi_j = \sum_{i \in S} \pi_i p_{ij}$

$$[\pi_1 \pi_2 \cdots \pi_j \cdots \pi_{|S|}] = \underbrace{[\quad]}_{P} \begin{matrix} \downarrow j\text{th coln} \\ \end{matrix}$$

The equation $\pi = \pi P$ (called **global balance equation**) is a system of $|S|$ linear equations. π is a left-eigenvector of P with eigen value 1.

From Chapman-Kolmogorov Equation $\pi(n+1) = \pi(1)P^n \forall n \in \mathbb{N}$

If $\pi(1) = \pi$ (the invariant distribution), $\pi(n) = \pi \forall n \in \mathbb{N}$.

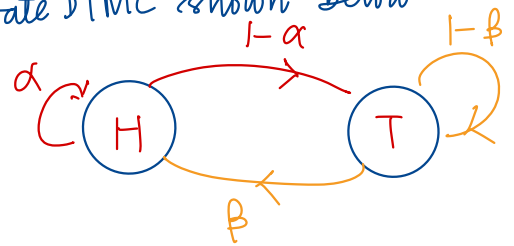
This, together with time-homogeneity implies that the Markov chain is **stationary**.

Homework: Find the stationary distribution of the two-state DTMC shown below.

$$\pi(0) = \alpha \pi(0) + \beta \pi(1)$$

$$\pi(1) = (1-\alpha)\pi(0) + (1-\beta)\pi(1)$$

Solve!



The invariant distribution may not be unique.

$$\pi_1 = \pi_1 P, \quad \pi_2 = \pi_2 P.$$

$$\pi_\alpha = \alpha \pi_1 + (1-\alpha) \pi_2, \quad \alpha \in [0,1]$$

$$\pi_\alpha = \pi_\alpha P.$$

Consider $P = I$

$$\pi = \pi P \quad \forall \pi$$

On existence and uniqueness of invariant distribution

Let $\{X_n, n \geq 0\}$ be an **irreducible**, time homogeneous DTMC on the state space S with TPM P . Then, a **unique stationary distribution** π exists if and only if $\{X_n, n \geq 0\}$ is **positive recurrent**.

In this case, $\pi_j = \frac{1}{v_{jj}} > 0 \quad \forall j \in S$ $\left\{ v_{jj} = \sum_{n=1}^{\infty} n f_{jj}^{(n)} \right\}$ — Mean recurrence time.

$f_{jj} = 1$ & $v_{jj} < \infty$ for a positive recurrent state j .